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GE Medical Systems

Technical Publications

IRIX Guide Version 1.0d



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IRIX Guide Version 1.0

Do not be afraid to try an old Sun UNIX command. Some of these still work in the SGI environment.

Remember that the SBC is still essentially an RP 2.X system. Use of commands from the UNIX Guide 3.0 that are designated as appropriate for the “HSARP” or “ALL”, or UNIX in general will work on the SBC. There is a single exception: **DO NOT force phase recovery on the SBC using the methods from the UNIX Guide!** It can be done, but only utilizing the methods in this document.

3.5 software has changed several procedures, and this list is not exhaustive. It should no longer be necessary to reset Informix for autofilming issues. cleanMon shuts the system down to the boot prompt. Follow the new procedure for halting applications only.

Conventions

Bold Face: Cross referenced links and system messages or prompts.

Italics: Variables you need to fill the real thing in for (e. g. *filename*).

Normal: Command syntax, purpose descriptions and paths.

No path - means that the executable is contained in the usr/g/bin directory and can be executed from anywhere. (login *useraccount*) indicates who you have to be to perform the command listed, when none is indicated, you can run the command from any user account.

User Accounts and Passwords

User Account	Password
genesis (SBC Only)	4\$app
insite	2getin
root	#bigguy
ctuser (OC Only)	4\$app
service	4rhelp

IRIX Operating Commands

Command	Purpose
!!	Repeat last command issued (see history).
!X	Repeat command indicated by number for X. (see history).
!X	Repeat last command issued from history list which begins with letter(s) indicated by X (see history).
(ALT) + F1	Bring a background window to the foreground.
(ALT) + F3	Push a foreground window into the background.
<CTRL>+c	Terminate current process.
<CTRL>+d	Logout of user account.
<CTRL>+z	End input.
alias	Display a list of all aliased commands.
arp -a	Display current arp (address resolution protocol) cache in memory. Will let you know who is in the current network routing table.
cd /directoryname/directoryname/ . . .	Change current working directory to that specified in /directoryname/directoryname/ . . .
chmod XXX filename	Change the permissions to XXX for filename. XXX is an octal number whose value determines read, write and executable status bits for a files permissions. drwxrwxrwx owneraccount size date filename - 111 111 111 ==> chmod 777 filename The first four bits are owner permissions, second are group permissions, and the third is everyone's permissions.
chown useraccount filename	Change filename ownership to useraccount.
cp /path/filename /path/filename	Copy the designated file to the destination indicated.
cu hostname	Establish full duplex serial connection with hostname, as in: cu sbc Commands supported by cu include: <enter>~. terminate connection <enter>~%b halt remote host
date MMddhhmmyy	Set the date and time as specified: MM is month, dd is day, hh is hour, mm is minute, yy is year. Must be root.
df -k	Display disk partition sizes. Will also display mounted file systems.
dkstat	Provide ongoing I/O statistics for all SCSI devices.
du	Display block size of current directory.
du -sk	Display block size of current directory and its contents.
eject SCSI Id	Ejects the SCSI Id specified.
find / -name "filename" -print	Locate filename and print its path searching the entire directory structure for which this account has permissions (login as root to catch them all).
fping Host	Send a 0 length packet to Host to check that Host is active on network.

IRIX Operating Commands

Command	Purpose
fsck	Run disk maintenance, answer yes to all questions it asks. If it asks questions, reboot when it is complete and repeat until it runs clean (login root). If system boots to single user UNIX (#): #cd /etc <enter> #fsck<enter> Output - questions are likely #reboot<enter> (see cd).
ftp <i>internet address</i>	File transfer protocol. Will connect to host designated by <i>internet address</i> . Login will be required. Most file system navigation will function normally. Other commands available include: • get <i>filename</i>-transfer <i>filename</i> from <i>internet address</i> to current directory on log host. • put <i>filename</i>-transfer <i>filename</i> to <i>internet address</i> from log host. • bye-logout of <i>internet address</i> and exit ftp. • ?-display all available commands
fx	Disk utility. Can be used to slip bad blocks. See Using fx to Slip Bad Blocks.
grep <i>text filename</i>	Search <i>filename</i> for any occurrences of <i>text</i> and print the line that contains it. Can also be used as a destination for a pipe as in: ps -ef grep <i>text</i>
history	Provides a list of all commands issued since login for current account. Use in conjunction with repeat command feature (!) to reissue same command.
kill <i>processid</i>	Terminate process indicated by <i>processid</i> (see ps -ef).
killall <i>processname</i>	Sends a shutdown signal to all software processes if no <i>process name</i> is specified or to the named process.
ln -s <i>destinationfilename filename</i>	Create a symbolic link between <i>destinationfilename</i> and <i>filename</i> . All output from <i>filename</i> will go to <i>destinationfilename</i> .
ls -alR	List current and all subdirectory contents.
mkdir <i>directoryname</i>	Create <i>directoryname</i> as a branch from the current directory.
more <i>textfilename</i>	Show the contents of <i>textfilename</i> a page at a time. Can also act as the destination of a pipe as in: tail -100 SYSLOG more.
mount <i>device directory</i>	This will display a table of mounted file systems with no <i>device directory</i> designation. By specifying a <i>device</i> and <i>directory</i> name, you can create a virtual partition that will allow you to work with the media in the <i>device</i> . (see umount)
netsnoop	Watch network activity.
netstat -ain	View network performance statistics. Collision rates from 3 to 5% will cause system performance degradation.

IRIX Operating Commands

Command	Purpose
netstat -r	View physically connected hosts on your net and the current network routing tables. Destination - direct connection (no router) Gateway - gate to destination Flags: • U - up • H - host • G - gateway • Note: current log host will show no flags. refcnt - current number of active uses per route Use - number of pacjets sent per route
nvrnm	Display current prom monitor settings.
ping <i>Host</i>	Function is similar to spray. Send a 64 byte packet to <i>Host</i> and have it return that packet. Provide statistics on status of transimission.
prtvtoc	Print out the root disk label.
prtvtoc /dev/rdisk/dksod[<i>n</i>]s0	Display partitions, sizes and SCSI controller information for SCSI disk. Must be root. (<i>[n]</i> is the SCSI device number).
ps -ef	Display running software processes.
ps -ef grep `whoami` more	Display all processes your current user account owns that are running.
pwd	Display current directory path.
rdate <i>suite_host</i>	Sync CPU time/date with that of <i>suite_host</i> .
reboot	Shutdown and restart IRIX.
rm -r <i>directoryname</i>	Delete directory contents and the <i>directoryname</i> .
rm <i>filename</i>	Delete <i>filename</i> .
rsh <i>suite_host</i>	Remote shell as current user to <i>suite_host</i> .
spray <i>Host</i>	Send 100 packets of length 100 to <i>Host</i> . Check that <i>Host</i> received all packets. Does not work for scanner hardware controllers.
sum <i>filename</i>	This command will generate a checksum value for <i>filename</i> . Use it to verify that the file you have remote copied or transmitted is the same as the original. Sum will produce the same result if it is.
swap -s	View available system swap space.
tail -f <i>textfilename</i>	Displays an ongoing update of <i>textfilename</i> . Useful for observing system startup as in: tail -f gesys_<suiteid>_OC0.log
tail -X <i>textfilename</i>	Show the last <i>X</i> lines of <i>textfilename</i> .
telnet <i>internetaddress</i>	Attempt to log into remote host designated by <i>internetaddress</i> .
top -i <i>second</i>	Display processes having the highest CPU usage updating every <i>second</i> .
touch <i>filename</i>	Create <i>filename</i> of zero length.
umount <i>filesystemname or directoryname</i>	This command will release a directory or file system. Use it to release a 3rd party mount chewing that is virtually chewing up root partition space. See mount and df .
w	What user accounts are active and what actions are they performing. Will show how long the ctuser c-shell has been up.
whoami	Print current userid.

CT/i Log Files

Log File	Host	Path	Description
.currentboards	SBC	/usr/g/service/log	This file contains the board revision levels for all boards serviced by the ETC, OBC, and STC. View it using the more command.
analysis.0	OC	/var/adm/crash	Provides a summary of all of the CPU activity, failure reason and executable involved in a CPU Panic. View it using more .
aqslog 	OC	/usr/g/ctuser/logfiles	Acquisition server log file. View it using tail or more .
arslog	OC	/usr/g/ctuser/logfiles	Archive server log file. View it using tail or more .
browserlog	OC	/usr/g/ctuser/logfiles	Browser client log file. View it using tail or more .
ctNum.cfg	OC	/usr/g/config	Current CT Number configuration file. View it using tail or more .
dbrlog	OC	/usr/g/ctuser/logfiles	Image database read server log file. View it using tail or more .
dbwlog	OC	/usr/g/ctuser/logfiles	Image database write server log file. View it using tail or more .
dcslog	OC	/usr/g/ctuser/logfiles	Dicom server log file. View it using tail or more .
dks1d1s[n]	OC	/etc/fscklogs	Contains the results of the fsck performed at startup for SCSI device indicated by [n]. (see CT/i Devices).
gesys_<suite>_OC0.log	OC	/usr/g/service/log	This is the same log you would get at in applications by hitting the message bar. Examine it at IRIX level using the viewlog command. Just about everything reports here, so it is the logical place to start troubleshooting.
hlfc.tapeRev	SBC	/login: root	List the SBC software revision. View using more . (see showprods)
Host.cfg	OC	/usr/g/config	Host configuration file. Use more to view this file. Not all parameters are used from this file.

CT/i Log Files

Log File	Host	Path	Description
imslog	OC	/usr/g/ctuser/logfiles	Image server log file. View it using tail or more .
INFO	OC	/usr/g/config	This file contains the system environment information. Use more to examine its contents to verify your answers to the installation procedure in reconfig .
messages messages breakpoints: messages.0, messages.1, etc.	SBC	/usr/adm	This is where UNIX hardware and operating system software errors are reported. Here you will find the reasons for CPU malfunctions, SCSI errors, RP faults and indications of system start ups. Use more or tail to view this text file or dmesg to look at the most current entry in the file.
netlog	OC	/usr/g/ctuser/logfiles	Network server log file. View it using tail or more .
prslog	OC	/usr/g/ctuser/logfiles	Print server log file. View it using tail or more .
ScanHardware.cfg	OC	/usr/g/config	Contains hardware configuration information supplied by you during the software LFC and by the system during initial polling of devices in the system. View it using more . Use grep to search for specific parameters. grep scanDiskDevice ScanHardware.cfg
scanUsageLog	SBC	/usr/g/service/log	Contains technique and scan types used on system since last software LFC. Use scanReport to view it.

CT/i Log Files

Log File	Host	Path	Description
 SdCRhosts	OC	/usr/g/ctuser/Prefs	This file contains all of of the hosts adopted at applications. Use it to find the IP addresses of other hosts you can telnet to, like AWWs.
SYSLOG	OC	/var/adm	This is where IRIX hardware and operating system software errors are reported. Here you will find the reasons for CPU malfunctions, SCSI errors, and indications of system start ups. Use more or tail to view this text file.
syslog syslog breakpoints: syslog.1, syslog.2, etc.	SBC	/usr/spool/log	Provides results of database regeneration. Can also provide clues to database errors. View it with more .

CT/i Scripts and Executables

Script/Executable	Host	Path	Purpose
ChangeTube	OC	No Path login: ctuser	Change the tube information.
cleanMon	OC	No Path login: ctuser	Shut applications and IRIX down. See shutdown procedure for apps only.
dbcount	OC	No Path login: ctuser	Indicate number of exams and images in the image database.
endsession	OC	No Path	Logout of current desktop and return to the login window.
ermes	OC	/usr/g/db login: ctuser	Rebuild corrupted gesys_<suite>_OC0.log.
exitScript	OC	No Path login: ctuser	Checks the status of applications and executes endsession if apps are down.
fwrev	SBC	No Path	List the firmware revision of all SCSI devices.
gattention <i>message</i>	OC	No Path	Put a gattention box on the monitor containing the text in <i>message</i> .
geacroread	OC	No Path login: ctuser	Launch Adobe Acrobat Reader.
halt	OC	No Path login: root	Shut operating system software down.
haltSystem	SBC	No Path	Shut operating system software down.
hard_conf	OC	No Path login: ctuser	Provides IRIX version, IP address, Ethernet address and other hardware CPU config info. For more information, use the hinv command
hinv	OC	No Path	Display hardware configuration information. For more information, use the hard_Conf command.
insight	OC	No Path login: root	Online SGI manuals. (graphical).
listselect	OC	/usr/g/insite/bin	Used to extract images from the image database prior to transmission over the MODEM to Insite. Images are placed in: /usr/g/insite/tmp Use ftp to transfer them once extracted.
memc	OC	/usr/tests/bin login: root	Test character memory.
memt	OC	/usr/tests/bin login: root	Test texture memory.

CT/i Scripts and Executables

Script/Executable	Host	Path	Purpose
mkfsMOD	OC	No Path login: root	Create a new file system on the MaxOptics disk side currently installed in the drive. This will clear out everything on the disk.
mountMOD	OC	No Path	Make the MaxOptics a virtual partition on the disk drive. Normal file functions to the /MOD directory (like cp, cd, ls, etc) will now function on the optical drive.
reconfig	OC, SBC	No Path login: root	Change site info, regen databases, etc. (must also be run on SBC once complete on OC). Use ChangeTube to install a new tube.
rhapSnap	OC	No Path	Store all logs to the MOD for later review. Run to assist with resolution of software issues.
rpbrdinfo	SBC	/usr/g/service/log login: genesis	Lists the boards installed in the RP and run an fwrev on the SCSI devices.
RunTimeStats	OC	No Path login: ctuser	Displays rotor time, motor currents, tube spits, RCOMM/SCOMM statistics.

CT/i Scripts and Executables

Script/Executable	Host	Path	Purpose
scanReport	SBC	No Path login: genesis	Displays contents of scanUsageLog .
showdasm	OC	No Path Path: ctuser	This will query the dasm and provide you with configuration information for it.
showprods grep Rhap	OC	No Path	List the revision of the system software. (see hlfc.tapeRev)
startMon	OC	No Path login: ctuser	Start applications processes.
start_SMPTE	OC	No Path login: ctuser	Installs the SMPTE pattern in the browser.
storelog	OC	No Path login: ctuser	Remove system log files from OC and SBC. Provides an option to store them to MOD. Reboot is required to complete the removal of the SYSLOG.
testethernet	OC	/usr/tests/bin login: root	Reports ethernet board status.
TubeUsage	OC	No Path login: ctuser	Displays the tube slice count for the last five tubes installed. Add the patient and non patient scans to obtain the total slice count.
unmountMOD	OC	No Path	Release the MaxOptics MOD's virtual partition.
unlockAll	SBC	No Path login: genesis	Frees stuck raw data headers.
viewlog	OC	/usr/g/insite/bin login: insite	Display gesys_<suite>_OC0.log with viewing controls.
whatRevEx <i>filename</i>	SBC	No Path	Will give the current revision of <i>filename</i> . Filename must be an executable.

CT/i Software Magic

Procedure	Host	Path	Purpose
hdrSrvTest	SBC	No Path login: genesis	Can be used to determine number of scan files in use and reserved. Perform following steps: <<menu list>> Enter Option->22 <enter> Enter suite . . .: 1 <enter> 1<enter> 1<enter> 1<enter> System will then display status of all raw data slices on system. Use this to verify rewritability of raw data slices. This utility is also capable of cleanups, reserve/release functions, etc. USE AT YOUR OWN RISK!
slay -f -1 dataacq	SBC	No Path login: genesis	Force the FEP to reset and reload its firmware.
slay -f -HUP fwmgr	SBC	No Path login: genesis	This will cause the scanner hardware to reset.
cd /usr/g/service/log<snapdirectoryname> tar -cvf ../hospitalsnap.tar . * * cd .. sum hospitalsnap.tar mountMOD cp hospitalsnap.tar /MOD cd /MOD sum hospitalsnap.tar compare the results of the sum commands. If equal, then: cd /usr/g/service/log rm hospitalsnap.tar rm -r <snapdirectoryname> unmountMOD eject and Fed Ex or UPS MOD to: Mr. Jim Nugent 16800 W. Ryerson Rd. New Berlin, WI 53151 Write your CQA. else: repeat from: cp hospital	OC	Path as given login: ctuser	This will allow you to copy the snap directory to the Max Optics MOD and verify the files. Replace <i>hospital</i> with your hopsital name. See also: cd, mountMOD, unmountMOD, sum, cp, rm.
grep sbc0 /etc/hosts	OC	Path as shown	This will display the SBC alias. See also: grep.

CT/i Software Magic

Procedure	Host	Path	Purpose
As ctuser: ctuser> ln -sf /sbin/echo /usr/g/bin/autoShutdown ctuser> cleanMon ctuser> top <i>(this will post the most active processes to the screen. Watch for ctuser to disappear, except for your top command)</i> rsh <suite>_sbc0 haltSystem y <enter> (connection closed) As root: reboot As ctuser: startMon <enter> <enter> <i>(startup messages)</i>	As Shown	No Path login: as shown	Shut down the system software and reboot IRIX and applications. See also: top, rsh, haltSystem, reboot, startMon, cleanMon.
Connect to system via telnet tool: <suite>%> mkdir dummyfiles open a terminal window: cd <directory of dummy files on your system> ftp <ip address of scanner> login: insite password: 2getin ftp>cd dummyfiles ftp> mput * <i>files get copied</i> ftp> bye The dummy files are now on the product and are ready to be replace the missing or zero length files. Make sure you login as root and set the correct file permissions and ownerships (chown, chmod) so that other executables can call these files as needed. Use ftp to copy them to the SBC, if required.	OC	Path: As Shown login: As Shown	This is the procedure to download small files to a product machine accessed via the PPP tool from the OnLine Center. The object is to allow a remote engineer to replace corrupted files on the system with good ones that will permit the scanning hardware to reset successfully. Once reset, the dummy files must be replaced with real files, by running the appropriate install program (e. g., install new tube). If onsite, the better solution is to simply restore system state. Dummy files can be obtained from GEEC or Ryerson systems. See also: ftp, mkdir, cd, chown, chmod.
ls -lags /dev/input/trakb* 1 lrwxr-xr-x 1 sys (date) /dev/input/trakb -> /dev/ttya3 ls -lags /dev/ttya3 0 crw-rw-rw- 1 sys (date) /dev/ttya3	OC	Path As Shown	Output should be as shown for proper operation of the trackball. See also: ls.

CT/i Software Magic

Procedure	Host	Path	Purpose
<i>shut down applications</i> <pre>cd /usr/g/scripts configTime -p ==> CST6CDT tail /etc/TIMEZONE <i>should match your entry above</i></pre> cd /usr/g/config vi INFO <i>Alter the fourth line in this file as:</i> setenv TZ CST6CDT reboot date <i>Mmddhhmm</i>yy rsh <suite>_SBC0 timezone <i>select proper timezone</i> reboot rsh <suite>_SBC0 rdate <suite>_OC0 exit <i>start applications</i>	As Shown	Path: As Shown login: root	This procedure will permit you to change the timezones and alter the date/times on the OC and SBC computers in order to get the time stamp on the images correct. The easiest way is to perform a reconfig and make the changes there. The values shown will set up central standard and daylight. The other USA choices are in the note below:  See also: vi, cleanMon, startMon, reboot, rsh, rdate, date.
<pre>ctuser> ln -sf /sbin/echo usr/g/bin/autoShutdown ctuser> cleanMon ctuser> rsh <suite>_sbc0 genesis> mkdir /usr/g/db/tmp genesis> cp /usr/g/db/genesis* /usr/g/db/tmp genesis> S root> hostMonitor & root> mrpm /R & root> rpTA & root> dbInit &</pre> <i>lots of messages begin, ending with:</i> <i>Recovery Successful. Delete old database.</i> <pre>root> slay -f dbserver root> slay -f dbInit root> slay -f mrpm root> slay -f hostMonitor root> haltSystem <i>Really . . . ? y</i> ctuser> su - root password: #bigguy root> reboot <i>system reboots</i></pre> ctuser> startMon	As Shown	As Shown	This will cause the SBC database to perform a non destructive phase recovery sequence. This can free stuck raw data headers. See also: rsh, mkdir, cp, haltSystem, reboot, cleanMon, startMon.

CT/i Software Magic

Procedure	Host	Path	Purpose
<i>Applications are UP to begin</i> OC: root> rsh <suite>_SBC0 SBC: root> cp /w/state/blocked_ref_info ~genesis/ blocked_ref_info root> exit OC: root> calConv -s /usr/g/service/log/FullCalDB root> exit ctuser> ln -sf /sbin/echo /usr/g/bin/autoShutdown ctuser> cleanMon SBC: ctuser> rsh <suite>_SBC0 genesis> resetSDB <i>answer 'y' (yes) to all questions</i> <i>Note: this is DESTRUCTIVE to the SBC database</i> genesis> exit OC: ctuser> startMon root> rsh <suite>_SBC0 SBC: root> cp ~genesis/ blocked_ref_info /w/state/blocked_ref_info root> exit root> calConv -r /usr/g/service/log/FullCalDB root> exit OC: ctuser> ln -sf /sbin/echo /usr/g/bin/autoShutdown ctuser> cleanMon ctuser> startMon	As Shown	As Shown	This will destructively reset the SBC databases and restore the databases. Try this if phase recovery does not work for your raw data issue. This will also work on any RP. See also: cp, rsh, cleanMon, startMon.

CT/i Software Magic

Procedure	Host	Path	Purpose
<i>Archive and film everything you can.</i> <i>From applications, have the customer remove as many images from the system as possible.</i> <i>shut down applications</i> ctuser> ln -sf /sbin/echo /usr/g/bin/autoShutdown ctuser> cleanMon ctuser> resetdatabase <i>This will take between 3 and 5 minutes to complete.</i> ctuser> su - root password: #bigguy #> reboot y <i>system reboots</i>	As Shown	As Shown	This will cause the image database to reset. All images will be deleted.
ctuser> ln -sf /sbin/echo /usr/g/bin/autoShutdown ctuser> cleanMon	OC	As shown	Execute these commands to shut applications down only.
/data	OC/SBC	n/a	This directory is mounted by the SBC and exists physically on the OC. If you place a file here, you will be able to grab it on the sbc by following the same "path."

vi Editor

Command	Effect
<i>vi filename</i>	Enter vi editor and load <i>filename</i> for editing.
h	Move cursor left one character. Cannot be in insert character mode; <ESC> exits this mode and returns to cursor control mode.
j	Move cursor down one line. Cannot be in insert character mode; <ESC> exits this mode and returns to cursor control mode.
k	Move cursor up one line. Cannot be in insert character mode; <ESC> exits this mode and returns to cursor control mode.
l	Move cursor right one character. Cannot be in insert character mode; <ESC> exits this mode and returns to cursor control mode.
i	Insert characters where the cursor is. Hit <ESC> to exit insert character mode.
a	Insert characters after the current cursor position. Hit <ESC> to exit insert character mode.
o	Insert characters on the next line. Hit <ESC> to exit insert character mode.
<ESC>:q!	Quit without saving changes.
<ESC><Shift>ZZ	Save changes and exit.
x	Delete current character.
dd	Delete current line.



CT/i Devices and Nomenclature

Device	IRIX Designation	Notes
4.2 GB System Disk	/dev/dsk/dksld1s [n]	[n] is the partition number.
4.2 GB Option Disk	/dev/dsk/dksld2s [n]	[n] is the partition number.
Max Optics MOD	/dev/dsk/dksld3s7	Use mountMOD and unmountMOD to access the optical media in IRIX.
Pioneer MOD	/dev/scsi/scld5l0	
CDROM	/dev/scsi/scld6l0	
DASM	/dev/scsi/sc0d1l0	SCSI channel 0.
keyboard	/dev/keybd	
mouse	/dev/mouse	PS/2 mouse
serial port 1	/dev/console	
serial port 2	/dev/ttyd2	Service Key Port
ethernet	ec0	
printer	/dev/cent	Centronics parallel printer port.
audio	/dev/audio	
Mardis Gras 1,0	/dev/gfx	Left CRT
Mardis Gras 1,1	/dev/gfx	Right CRT
GIO64 Bit3	vd0	
Specialix serial ports:	/dev/ttya1-ttya8 /dev/ttya1 /dev/ttya2 /dev/ttya3 /dev/ttya4 /dev/ttya5 /dev/ttya6-8	serial expansion ports link to SBC console port touch panel input trackball input port for laptop ^(not supported) Dynaplan link to STC serial #2 Not used



Using fx to Slip Bad Blocks

Please note that fx can be very dangerous to disk data and it can easily destroy system software!

Bad blocks can be detected and reported by fsck during a reboot as a "medium error." Two different block locations are reported by fsck: the first number is the bad block relative to the disk partition it was found in; the second number, in parenthesis, is the bad block relative to the entire disk.

Bad blocks can also be reported to the SYSLOG or as runtime panics with a system disk "medium error" posted to the screen. The details may vary slightly but the concept is the same (record bad block#, use fx to slip it).

To repair and boot the system, it is necessary to boot the **fx** disk utility from disk (or from CDROM if the disk boot of fx fails) with "boot /stand/fx" from the SGI firmware command monitor prompt.

When fx asks you for "special privileges" answer yes. Enter the correct device information (as shown below) and then go into the "badblock" menu and use the "addblock" function to enter the **SECOND** number reported by fsck (the one in parenthesis).

Example Procedure

```
fx version 5.3, Oct 9, 1995
```

```
fx: "device-name" =(dksc)
```

```
fx: ctrl# = (0) 1
```

```
fx: drive# = (1)
```

```
fx: lun# = (0)
```

```
...opening dksc(1,1,0)
```

```
...controller test...OK
```

```
Scsi drive type == SEAGATE ST15150N    0017
```

```
----- please choose one (? for help, .. to quit this menu)-----
```

```
[exi]t      [d]ebug/      [l]abel/      [a]uto  
[b]adblock/  [exe]rcise/    [r]epartition/  [f]ormat  
fx> b
```

```
----- please choose one (? for help, .. to quit this menu)-----
```

```
[a]ddbb      [s]howbb  
fx/badblock> a
```

```
please enter a bn from 0 to 8387666
```

```
fx/badblock/addbb: add badblock =123456 <==use your block number
```

```
trying to save the data ...adding bad block (123456)
```

```
rewrote saved data OK
```

```
please enter a bn from 0 to 8387666
```

```
fx/badblock/addbb: add badblock= (USE CONTROL'C' AT THIS POINT TO EXIT)
```

```
----- please choose one (? for help, .. to quit this menu)-----
```

```
[a]ddbb      [s]howbb  
fx/badblock> ../exi
```

It is very likely that the "trying to save the data" portion above will fail and retry many times but fx will eventually just slip the bad block anyway. This means there will be corrupt data on the disk which may or may not be a problem since there are thousands of unused/unnecessary files on Unix/Irix systems. In the worst case, an LFC may be necessary after the bad block repair.